

A Branch A cohomological Higgs module: stabilizer-derived quartic overlap and lightness-protection gates

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Abstract

We describe a constrained Branch A projection/cohomology Higgs module in which the observed four-dimensional Higgs is treated as the visible projection of a tau-profiled parent field. The central reproducible calculation links a minimal $3 + 2$ stabilizer, the canonical hypercharge direction, a Higgs localization exponent $\nu_D = 3/10$, and a zero-mode quartic overlap $I_4(3/10) = 0.133756$. This is a concrete conditional mathematical result: under the stated Branch A rule, the observed Higgs quartic corresponds to an order-one parent quartic rather than to an extreme hierarchy. The construction is a candidate mechanism with explicit validation gates. It is not a completed Standard Model derivation, not a proof of any parent projection theory, and not an empirical claim.

1 Scope and Claim Boundary

This manuscript isolates one theoretical module: a possible origin for a Higgs-scale quartic overlap and a possible cohomological lightness-protection route. The paper does not claim that the Standard Model, gravity, or a complete parent projection theory has been derived. The result should be read as a reproducible candidate mechanism with explicit proof obligations.

2 Minimal Projection Setup

The tau coordinate is treated as an internal projection coordinate rather than an ordinary fifth spacetime dimension. A parent Higgs profile is written as

$$H(x^\mu, x), \tag{1}$$

where x^μ is ordinary four-dimensional spacetime and x is the internal tau/projection coordinate. The observed Higgs is the visible projection

$$\phi(x^\mu) = \Pi_{\text{vis}}[H] = \int h_D(x) H(x^\mu, x) dx. \tag{2}$$

Projection-null directions have the form $Q_D^\dagger \Lambda$ and are proposed to be quotiented out:

$$H \sim H + Q_D^\dagger \Lambda. \tag{3}$$

3 Branch A Stabilizer

Let the parent internal space split as

$$V \simeq \mathbb{C}^5 = C \oplus W, \quad \dim C = 3, \quad \dim W = 2. \quad (4)$$

The traceless two-block generator is

$$T_\Sigma = \frac{1}{\sqrt{60}} \text{diag}(2, 2, 2, -3, -3). \quad (5)$$

Its stabilizer is $S(U(3) \times U(2))$, giving the Standard-Model-compatible non-Abelian stabilizer structure. With

$$Y = \text{diag} \left(-\frac{1}{3}, -\frac{1}{3}, -\frac{1}{3}, \frac{1}{2}, \frac{1}{2} \right), \quad (6)$$

the canonically normalized hypercharge generator satisfies $T_\Sigma = -T_Y$. Equivalently,

$$T_Y = \sqrt{\frac{3}{5}} Y, \quad (7)$$

so the Branch A stabilizer fixes the familiar hypercharge-normalization factor $\kappa_\tau^2 = 3/5$ used below. This numerical factor is therefore not fitted to the Higgs quartic; it is inherited from the stabilizer-compatible normalization of the hypercharge direction.

4 Higgs Exponent and Zero Mode

The Branch A localization postulate is treated here as an assumption/theorem-candidate:

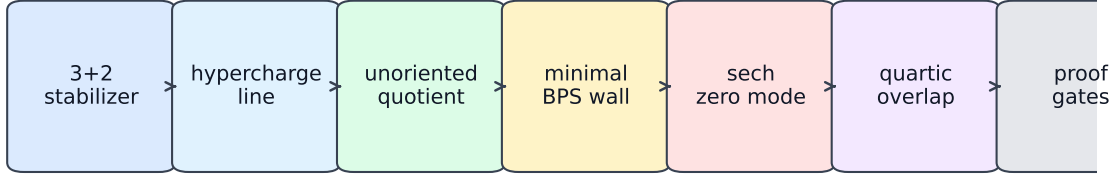
$$\nu_i = \kappa_\tau^2 |Y_i| = \frac{3}{5} |Y_i|. \quad (8)$$

This separation is important. The factor $3/5$ follows from the Branch A hypercharge normalization above; the main theoretical blocker is the remaining localization postulate $\nu_i = \kappa_\tau^2 |Y_i|$. That postulate must eventually be derived from a stabilizer-compatible metric, a variational extremum, an index-theoretic constraint, anomaly matching, or another independent principle. Until such a derivation exists, the quartic-overlap result should not be read as a derivation of the Higgs quartic.

The limited claim tested here is narrower but meaningful. Among linear hypercharge-localization rules $\nu_i = c |Y_i|$, the value $c = 3/5$ is the Branch A working value that keeps the Higgs mode localized and gives an order-one parent quartic requirement. This establishes the quantitative target that the derivation gate must explain; it does not close that gate by itself.

5 Theorem Status Summary

Claim	Status in this manuscript
$3 + 2$ stabilizer	derived within the minimal Branch A carrier
$\kappa_\tau^2 = 3/5$ normalization	derived from canonical hypercharge normalization
unique Abelian direction	derived within the traceless $3 + 2$ Branch A centralizer
unoriented T_Y line quotient	theorem-candidate, with projector audit
$\tanh x$ wall	candidate minimal/BPS wall route
$\nu_i = \kappa_\tau^2 Y_i $	theorem-candidate, not yet parent-derived
quartic overlap $I_4(3/10)$	computed and independently audited
projection-BRST protection	roadmap gate
top determinant / radiative stability	open gate



conditional route: derived pieces feed theorem-candidate gates before physical claims

Figure 1: Mechanism route and proof-gate structure. The first steps are controlled algebraic reductions; the localization rule and physical interpretation remain conditional on the parent-action gates.

6 Candidate Route To The Localization Rule

The localization postulate can be sharpened into a candidate derivation route. Assume that the tau-direction zero-mode operator for a field component i is a covariant first-order operator

$$Q_i = \partial_x + A_{\tau,i}(x), \quad (9)$$

and that the Branch A stabilizer induces a hypercharge-directed tau connection

$$A_{\tau,i}(x) = \kappa_\tau^2 Y_i \tanh x. \quad (10)$$

Then the chiral zero-mode equation

$$Q_i h_i = 0 \quad (11)$$

has the local solution

$$h_i(x) \propto \exp \left[- \int^x \kappa_\tau^2 Y_i \tanh u \, du \right] = \text{sech}^{\kappa_\tau^2 Y_i} x. \quad (12)$$

Normalizability selects the sign/chirality of the localized branch, so the positive localization exponent is

$$\nu_i = |\kappa_\tau^2 Y_i| = \kappa_\tau^2 |Y_i|. \quad (13)$$

This is not yet a complete proof, because the hypercharge-directed connection $A_{\tau,i}(x) = \kappa_\tau^2 Y_i \tanh x$ must itself be derived from the parent metric, connection, index problem, or BRST quotient. It does, however, isolate the missing theorem: once the tau connection is fixed by the canonically normalized hypercharge generator, the zero-mode exponent follows algebraically.

7 Stabilizer Origin Of The Hypercharge Direction

Branch A starts from the $3 + 2$ split

$$V = C \oplus W, \quad \dim C = 3, \quad \dim W = 2. \quad (14)$$

A connection that preserves the non-Abelian stabilizer $SU(3) \times SU(2)$ must be block-scalar on C and on W :

$$T = \text{diag}(a, a, a, b, b). \quad (15)$$

The traceless condition gives

$$3a + 2b = 0. \quad (16)$$

Thus, up to an overall normalization, there is only one Abelian generator that commutes with the Branch A non-Abelian stabilizer. Choosing the conventional Standard Model hypercharge signs gives

$$Y = \text{diag} \left(-\frac{1}{3}, -\frac{1}{3}, -\frac{1}{3}, \frac{1}{2}, \frac{1}{2} \right). \quad (17)$$

After canonical normalization this is the same direction as $-T_\Sigma$. Therefore a minimal stabilizer-preserving tau connection has no independent Abelian direction available other than hypercharge:

$$A_\tau(x) = f(x) T_Y. \quad (18)$$

This is the sense in which the hypercharge alignment is derived from the Branch A stabilizer. A stronger proof would still have to show that the parent connection dynamically chooses this minimal stabilizer-preserving Abelian direction rather than a larger symmetry-breaking sector.

8 Minimal Wall Profile And Quotient Vacua

Once the direction is fixed, the remaining question is the scalar interpolation $f(x)$. The minimal domain-wall ansatz assumes a smooth odd profile connecting two constant asymptotic tau vacua:

$$f(-\infty) = -1, \quad f(+\infty) = 1, \quad f(0) = 0. \quad (19)$$

The standard first-order kink equation

$$f'(x) = 1 - f(x)^2 \quad (20)$$

has the unique centered solution

$$f(x) = \tanh x. \quad (21)$$

Equivalently, this is the Bogomolny equation for the quartic domain-wall potential

$$V(f) = \frac{1}{2}(1 - f^2)^2, \quad (22)$$

after choosing the tau length scale so that the wall width is one. The same choice also yields the Pöschl–Teller supersymmetric zero-mode operator

$$Q_i = \partial_x + \kappa_\tau^2 Y_i \tanh x, \quad (23)$$

whose normalizable solutions are powers of $\text{sech } x$.

Thus the $\tanh x$ profile is not selected to fit the Higgs quartic. It is the minimal centered kink profile for a smooth two-vacuum tau wall. What remains open is whether the parent projection geometry or quotient action forces this minimal kink rather than allowing a more general monotone profile.

The two-vacuum part of this wall route can be given a sharper quotient interpretation. The Branch A Abelian direction is a one-dimensional internal line

$$L_Y = \text{span}(T_Y), \quad (24)$$

not necessarily a signed observable vector. If the parent regular background is the unoriented line, or equivalently the rank-one projector onto that line, then the two oriented representatives are identified:

$$\text{span}(T_Y) = \text{span}(-T_Y), \quad P_Y = P_{-Y}. \quad (25)$$

Thus the quotient identifies

$$\Sigma \sim -\Sigma. \quad (26)$$

For $\Sigma = fT_Y$, this gives

$$f \sim -f. \quad (27)$$

The regular invariants are sign-even:

$$\text{Tr}(\Sigma^2) = \frac{1}{2}f^2, \quad \text{Tr}(\Sigma'^2) = \frac{1}{2}(f')^2. \quad (28)$$

Therefore odd powers of f would not be well defined on the quotient. If the regular branch fixes the line magnitude,

$$2 \text{Tr}(\Sigma^2) = 1, \quad (29)$$

then $f^2 = 1$, giving the two oriented representatives $f = \pm 1$. In this sense, the two vacua are not two unrelated signed scalar states; they are the two orientations of the same internal Branch A line.

This closes the two-vacuum logic conditional on a parent theorem that the Branch A regular datum is an unoriented tau-normal line or projector. If the full parent theory instead treats $+T_Y$ and $-T_Y$ as distinct observable charges, this quotient argument fails and the two-vacuum wall must be supplied by a separate dynamical mechanism.

For the Higgs doublet, $Y_H = 1/2$, giving

$$\nu_D = \frac{3}{5} \cdot \frac{1}{2} = \frac{3}{10}. \quad (30)$$

The first-order operator is

$$Q_D = \partial_x + \frac{3}{10} \tanh x, \quad (31)$$

and the visible zero mode solves $Q_D h_D = 0$:

$$h_D(x) = \mathcal{N}_D \operatorname{sech}^{3/10} x. \quad (32)$$

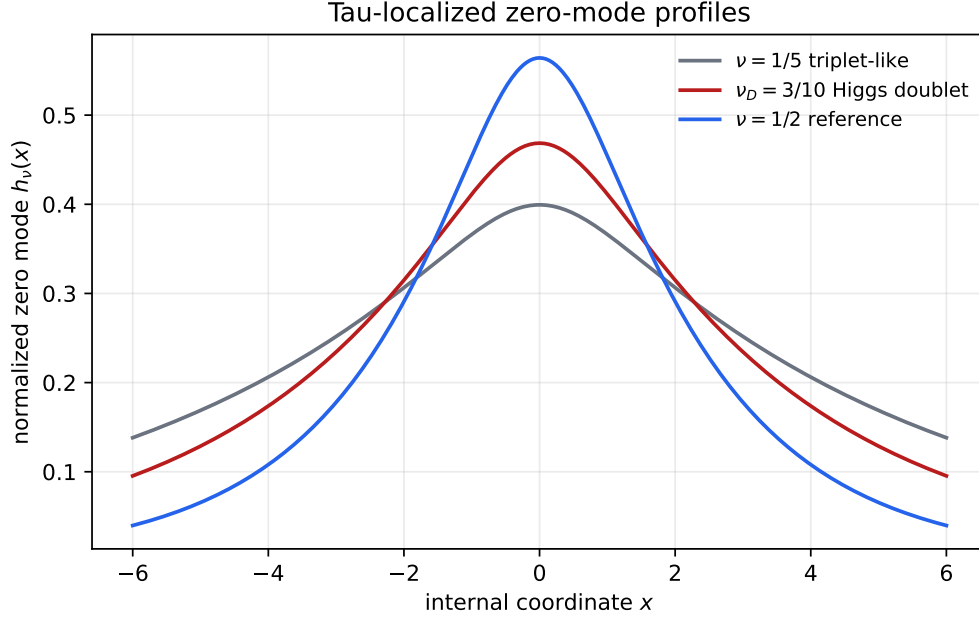


Figure 2: Normalized tau-localized zero-mode profiles. The Higgs doublet exponent $\nu_D = 3/10$ is compared with a triplet-like exponent and a reference exponent.

9 Quartic Overlap And Sensitivity Audit

For a normalized profile h_ν , the quartic localization functional is

$$I_4(\nu) = \int h_\nu^4 dx = \frac{\Gamma(\nu + 1/2)^2 \Gamma(2\nu)}{\sqrt{\pi} \Gamma(\nu)^2 \Gamma(2\nu + 1/2)}. \quad (33)$$

For $\nu_D = 3/10$ this gives

$$I_4(3/10) = 0.133756. \quad (34)$$

If the parent quartic is $\lambda_\tau(H^\dagger H)^2$, then

$$\lambda_H = \lambda_\tau I_4(3/10). \quad (35)$$

For $\lambda_H \simeq 0.129$, the implied parent quartic is

$$\lambda_\tau \simeq 0.964. \quad (36)$$

The important point is not the exact closeness to one. The safer statement is that the overlap requires an order-one parent quartic rather than an extreme hierarchy. The canonical parent quartic is not derived in this manuscript.

To reduce the numerology risk, the generator writes an explicit sensitivity table. In the moderate window $0.25 \leq \nu \leq 0.35$, the required parent quartic ranges approximately from 0.847 to 1.129. This means the mechanism is not a delta-function coincidence at $\nu = 0.3$: a neighborhood of nearby exponents still maps the observed quartic to an order-one parent coupling. The sensitivity audit strengthens the mechanism as a robust target, while still not proving the hypercharge-localization rule.

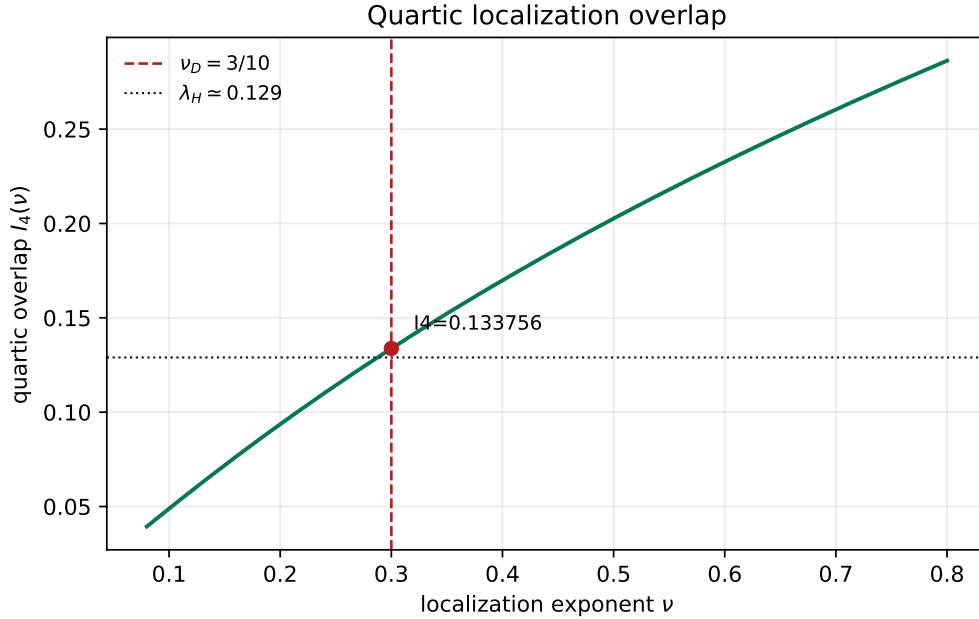


Figure 3: Quartic localization overlap as a function of the tau-localization exponent. The Branch A Higgs value $\nu_D = 3/10$ is marked, but the figure should be read as a sensitivity audit rather than as standalone evidence.

10 Cohomological Protection

The quotient $H \sim H + Q_D^\dagger \Lambda$ suggests a projection-BRST implementation,

$$sH = Q_D^\dagger c, \quad sc = 0, \quad s\bar{c} = b, \quad sb = 0. \quad (37)$$

At the exact cohomological point the critical operator is factorized,

$$\mathcal{O}_0 = Q_D^\dagger Q_D, \quad (38)$$

so the visible zero mode is massless. A local parent mass $\int H^\dagger H dx$ does not descend to the quotient because it is not invariant under the vertical redundancy. This is only a possible lightness-protection route. The current manuscript does not solve the Higgs hierarchy problem. A complete treatment would require the Hilbert-space domain, nilpotency, regulator, anomaly, and Ward-identity analysis. Here the projection-BRST language is retained as a roadmap gate, not as a completed QFT closure proof.

11 Controlled Top/Flavor Deformation

The leading cohomology-breaking deformation is written as

$$\mathcal{O}_D = Q_D^\dagger Q_D - \delta_\star W_D, \quad W_D(x) = \frac{3}{10} \text{sech}^2 x. \quad (39)$$

The zero-mode expectation value gives

$$\int h_D^2 W_D dx = \frac{9}{80}, \quad (40)$$

and therefore

$$\mu_H^2 = \frac{9}{80} \delta_\star M_\tau^2. \quad (41)$$

This section is a roadmap calculation: the top determinant must still be completed before the deformation can be promoted to a derived prediction. In particular, the manuscript does not yet prove radiative stability or solve the hierarchy problem. The top/flux determinant is the hardest remaining gate: it must show that mismatch-independent and linear mass terms are absent or quotient-trivial.

12 What Is Reproduced And What Is Not

The current package reproduces:

- a minimal $3 + 2$ stabilizer-compatible hypercharge direction;
- the Branch A working exponent $\nu_D = 3/10$ once the $\nu_i = 3|Y_i|/5$ rule is assumed;
- the normalized $\text{sech}^{3/10}$ zero mode;
- the quartic overlap $I_4(3/10) = 0.133756$;
- an order-one parent-quartic requirement and a TeV-scale deformation estimate.

It does not reproduce:

- the full Standard Model;
- the Yukawa hierarchy;
- anomaly cancellation from the parent construction;
- radiative stability;
- the measured Higgs mass from a completed top determinant;
- a collider-ready heavy-sector spectrum.

13 Wolfram Language Audit

The repository includes optional Wolfram Language audit scripts as an independent symbolic/numeric check of the formal skeleton:

- `Higgs_Quartic_Overlap_Verification.wl` checks the normalized $\text{sech}^\nu x$ profile, the quartic-overlap formula, $I_4(3/10)$, and the sensitivity range for the required parent quartic;
- `BranchA_Stabilizer_Hypercharge_Audit.wl` checks $\text{Tr } T_\Sigma = 0$, $\text{Tr } T_\Sigma^2 = 1/2$, $\text{Tr } T_Y^2 = 1/2$, and $T_\Sigma = -T_Y$, thereby verifying the stabilizer and hypercharge-normalization origin of the $3/5$ factor;
- `G2_Unoriented_Line_Quotient_Audit.wl` checks that the line projector for T_Y is invariant under $T_Y \mapsto -T_Y$, that $\text{Tr}[(fT_Y)^2] = f^2/2$, and that the unit-line condition gives the two oriented representatives $f = \pm 1$;
- `Projection_BRST_Skeleton.wl` checks only the algebraic skeleton $s^2 H = 0$ and $Q_D h_D = 0$.

The generated log files are included in the reproducibility packet. These checks support the formula and algebra audit. They verify the normalization source of $3/5$, but they do not derive the localization postulate $\nu_i = \kappa_\tau^2 |Y_i|$, prove anomaly freedom, prove regulator safety, or establish radiative stability.

14 Why This Is Not Just Numerology

The quartic-overlap calculation is evidence for internal coherence of the Branch A Higgs module, but not evidence for the full physical theory. The module becomes scientifically decisive only if the gates are passed in the right order: first derive the localization postulate that ties the exponent to $\kappa_\tau^2 |Y_i|$, then prove the quotient/anomaly safety, then complete the top determinant, and only then compare the resulting heavy-sector predictions to collider constraints. Without those gates, the value $I_4(3/10)$ should be treated as a reproducible mechanism target rather than as physical validation.

15 Remaining Proof Gates

The present manuscript isolates three unresolved gates. These are not presentation details; they are the conditions that separate the current mechanism target from a Higgs-sector derivation.

1. **Localization derivation.** The factor $3/5$ is now traced to the Branch A hypercharge normalization, but the localization postulate $\nu_i = \kappa_\tau^2 |Y_i|$ still has to be derived. A successful derivation would need to show that the Branch A regular background is the unoriented hypercharge line/projector, that non-hypercharge leakage modes are Hessian-positive or quotient-null, and that the reduced wall functional is the minimal BPS two-vacuum functional in the canonical tau measure.
2. **BRST/anomaly/regulator consistency.** The projection-BRST skeleton explains how a quotient could protect the visible zero mode, but the present packet checks only the algebraic skeleton. A proof must still define the Hilbert-space domain, nilpotent charge, regulator, anomaly constraints, and Ward identities.

3. **Top determinant and radiative stability.** The top/ﬂavor deformation is still a roadmap calculation. Until the determinant is computed and mismatch-independent or linear mass terms are shown to be absent or quotient-trivial, the manuscript does not solve the Higgs hierarchy problem.

These gates make the status of the paper precise. The quartic-overlap result is a reproducible and nontrivial mechanism target, but the localization, cohomological consistency, and radiative-stability gates must all close before the module can be promoted to a derivation.

16 Near-Term Falsifiable Prediction

The current module predicts a narrow kind of future test rather than an already validated signal. If the top/ﬂavor deformation gate is completed with a natural spurion $\delta_\star \sim 10^{-3}$, the same equations imply a parent scale M_τ in the multi-TeV range and a Higgs-sector spectral threshold

$$m_{\text{gap}} \simeq \nu_D M_\tau \sim 2\text{--}4 \text{ TeV}. \quad (42)$$

The associated Higgs-coupling deviations should be parametrically of order

$$\frac{\Delta g}{g} \sim \frac{v^2}{m_{\text{gap}}^2} \sim 10^{-3}\text{--}10^{-2}. \quad (43)$$

This is a falsifiable target, not a confirmed prediction. The module weakens if the completed determinant forces much larger deviations, no TeV-sector threshold, or collider-excluded states.

17 Validation Gates

The module would fail if any of the following gates fail:

- the Branch A projection metric rule $\nu_i = 3|Y_i|/5$ cannot be derived;
- the parent quartic is not canonical or requires large tuning;
- the projection-BRST quotient is anomalous or regulator-dependent;
- a visible Higgs mass is unavoidable at $\delta_\star = 0$;
- the top determinant produces a large mismatch-independent or linear mass term;
- the TeV-sector spectral window is excluded by Higgs coupling or direct-search constraints.

18 Conclusion

The Branch A Higgs module links a $3 + 2$ stabilizer, hypercharge normalization, a $\text{sech}^{3/10}$ visible zero mode, and a quartic overlap requiring an order-one parent quartic. The paper establishes a compact, reproducible mechanism target: if the Branch A localization rule is derived, the Higgs quartic is mapped to a natural parent-scale coupling rather than an extreme hierarchy. It does not establish the full parent theory or solve the Higgs hierarchy problem. The next paper-grade step is to derive the $\nu_i = 3|Y_i|/5$ rule, complete the projection-BRST and top-determinant gates, and compare the implied heavy-sector window against collider constraints.

References